

Medical Image Morphological Processing Using Brain MRI

Non-learning OpenCV pipeline for structural segmentation, binarization, and boundary extraction.



Dataset Source

Brain MRI Scan (Neg0.jpg)



Core Stack

Python • OpenCV • NumPy

Mathematical Morphology Objectives



Light Normalization

Standardized 256×256 grayscale matrix.



Otsu Binarization

Automated tissue-background segmentation.



Set Morphology

Erosion, Dilation, Opening, Closing.



Boundary Search

Internal contour via set difference.

Brain MRI Acquisition & Normalization

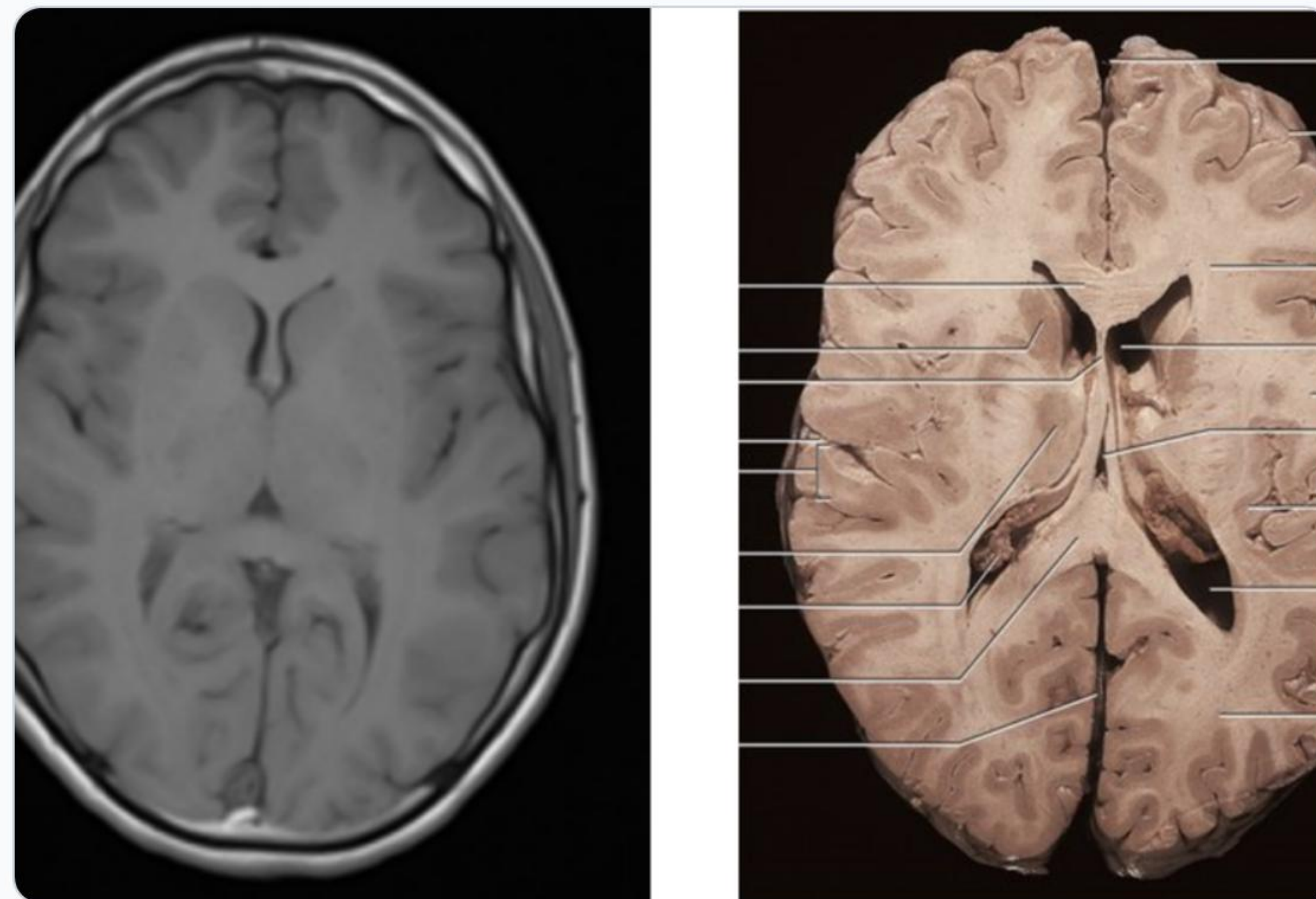
Automated Multi-Path Loader: Searches local repository and mounted Google Drive dynamically.

Single-Channel Conversion: Loads scan directly into 8-bit grayscale space (`cv2.IMREAD_GRAYSCALE`).

Spatial Resampling: Standardizes input to 256×256 matrix resolution.

Deterministic Pipeline: Eliminates machine learning training dependencies.

Output Dimension: 256 × 256 Pixel Matrix



Intensity Distribution & Contrast Metrics

MEAN INTENSITY (μ)

119.79

Average global brightness

STD DEVIATION (σ)

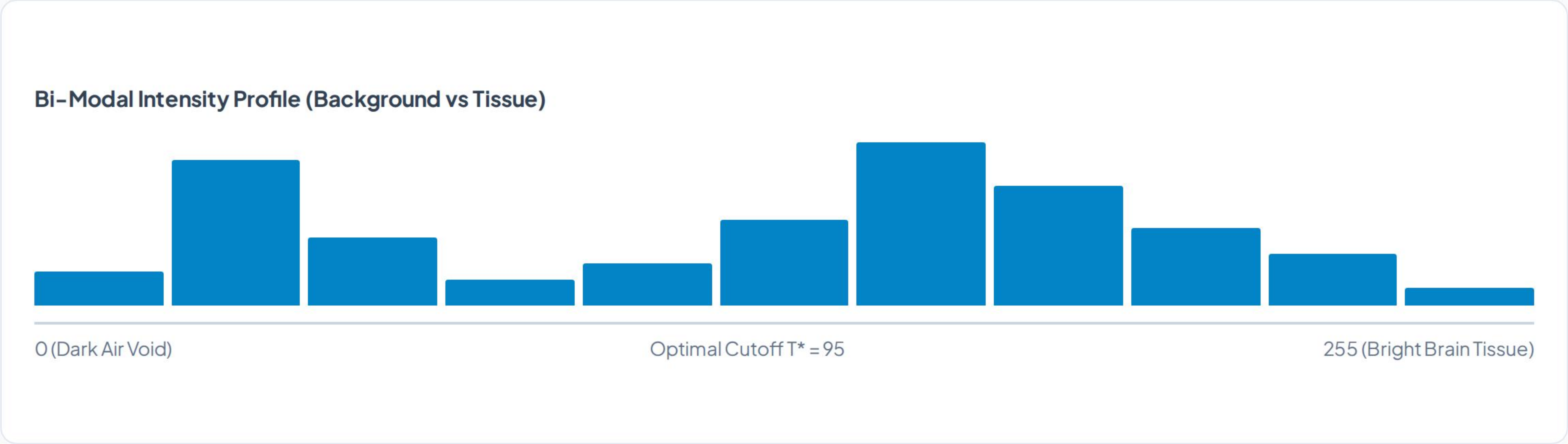
49.75

Intensity variance & contrast

DYNAMIC RANGE

[12, 255]

Full 8-bit scale utilization



Otsu Automatic Binarization

OPTIMAL THRESHOLD (T^*)

95

Calculated via intra-class variance minimization.

Non-Parametric Segmentation

No manual intensity tuning required.

Binary Matrix Generation

Converts image to boolean $\{0, 255\}$ state.

Tissue Separation

Isolates brain parenchyma from background air.

Primary Operators: Erosion vs. Dilation

Erosion ($A \ominus B$)

Shrinks

Mechanism: Minimum pixel value in kernel window.

Effect: Strips foreground boundaries.

Use Case: Removes small outer noise spurs.

```
OpenCV: cv2.erode(mask, kernel, iter=1)
```

Dilation ($A \oplus B$)

Expands

Mechanism: Maximum pixel value in kernel window.

Effect: Expands tissue region boundaries.

Use Case: Bridges narrow interior gaps.

```
OpenCV: cv2.dilate(mask, kernel, iter=1)
```


Compound Operators: Opening vs. Closing

Opening ($A \circ B$)

Erosion → Dilation

Sequence: Erosion followed immediately by Dilation.

Primary Function: Removes background noise and thin bridges.

Area Impact: Preserves overall object area and shape.

OpenCV: MORPH_OPEN

Closing ($A \bullet B$)

Dilation → Erosion

Sequence: Dilation followed immediately by Erosion.

Primary Function: Fills small internal tissue holes.

Area Impact: Smooths contours and joins channels.

OpenCV: MORPH_CLOSE

Morphological Boundary Extraction

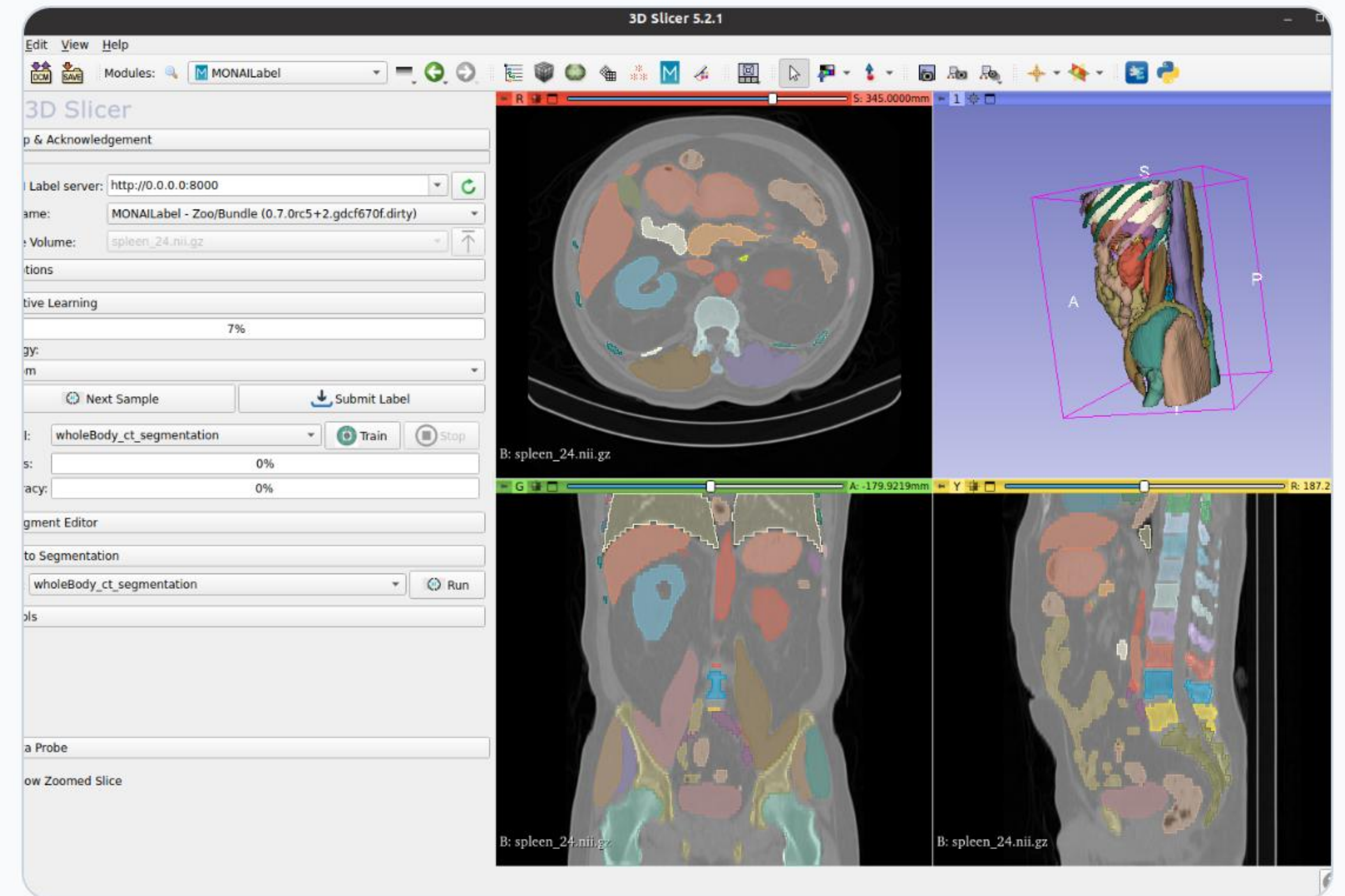
MATHEMATICAL FORMULA

$$\beta(A) = A - (A \ominus B)$$

Set Difference: Subtracts eroded mask from original binary mask.

Single-Pixel Contour: Produces exact 1-pixel structural outline.

Red Channel Overlay: Superimposes boundary onto grayscale scan.



Structuring Element Scale Impact

3 × 3 Kernel

Mild Thinning

Preserves fine anatomical details.

5 × 5 Kernel

Moderate Shrinking

Smooths structural outer edges.

7 × 7 Kernel

Severe Loss

Erodes critical thin structures.

Master Pipeline Execution Flow

1 Input & Normalization
Grayscale read and 256x256 matrix resize.

2 Otsu Thresholding
Auto-bimodal cutoff calculation ($T^* = 95$).

3 Morphological Filter Suite
Apply Erosion, Dilation, Opening, and Closing.

4 Boundary & Overlay Rendering
Extract contour difference and render red overlay.

Clinical & Computational Takeaways



Ultra-Fast Execution

Sub-millisecond processing per frame.



Pre-Segmentation Utility

Ideal preprocessing for deep learning.



Deterministic Precision

Zero training dataset requirements.

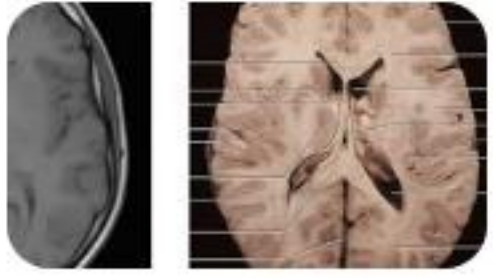
CONCLUSION

Questions & Discussion

Medical Image Morphological Processing Pipeline Complete

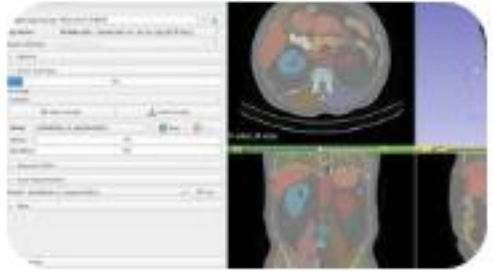


Image Sources



<https://airsmed.com/wp-content/uploads/2023/07/Brain-MRI-and-real-brain-1024x603.png>

Source: airsmed.com



<https://developer-blogs.nvidia.com/wp-content/uploads/2023/06/structure-segmentation.png>

Source: developer.nvidia.com